Difference Equations

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Introduction

While dealing with problems in Electrical and Electronics Engineering, specially in the area of digital signal processing and digital filter, we come across sequences of discrete signals. If we write a sequence in digital processing as $\{u_n\}$ then $u_n = nT$ where $n = 0, 1, 2, \ldots$ and T is called sampling period. The operation of such discrete systems is governed by difference equations which are often solved by Z-transforms.

What is a difference equation?

Definition (Difference Equation)

A difference equation is a relation between various differences of an unknown function at one or more values of the argument which is always non-negative.

Example

For example,

$$\Delta y_{n+1} + 3y_n = 2, \qquad \Delta^2 y_{n-1} + \Delta y_{n+1} = 5$$

are all difference equations where

$$\Delta y_{n+1} = y_{n+2} - y_{n+1} \tag{1}$$

and

$$\Delta^2 y_n = y_{n+2} - 2y_{n+1} + y_n \tag{2}$$

Thus, the above difference equations can be written respectively as

$$y_{n+2} - y_{n+1} + 3y_n = 2 \quad \text{and} \quad y_{n+2} - 2y_n + y_{n-1} = 5$$

Note

The difference equations are sometimes referred to as recurrence relations also.

Definition (Order of difference equations)

The order of a difference equation is defined as difference between the largest and the smallest arguments occurring in the difference equations, divided by the unit of increment. Thus,

$$\text{The order of a difference equation} = \frac{\text{largest argument} - \text{smallest argument}}{\text{unit of argument}}$$

Example

The order of the difference equation (1) is

$$\frac{n+2-n}{1} = 2$$

and the order of the difference equation (2) is

$$\frac{(n+2)-(n-1)}{1} = 3.$$

Similarly, the order of the difference equation $\Delta^2 y_{n+1} + 3\Delta y_n + 4y_n = 2$ is $\frac{n+3-n}{1} = 3.$

Definition (Solution of the difference equation)

An expression for y_n which satisfies given difference equation is called the solution of the given difference equation.

Formation of difference equation

Formation of difference equation from given expression is analogous to those of differential equation. We have to eliminate the arbitrary constants from given expression using different differences to obtain difference equation. How difference equations are formed , will be better explained through the following examples.

Example

Form the difference equation for the sequence $\{y_n\}$ where $y_n = a3^n + b5^n$, where a and b are arbitrary constants.

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Solution

$$y_n = a3^n + b5^n \tag{3}$$
$$\implies y_{n+1} = 3a3^n + 5b5^n \tag{4}$$
$$\implies y_{n+2} = 9a3^n + 25b5^n \tag{5}$$

Solving for a and b from (4) and (5) yields

$$a = \frac{5y_{n+1} - y_{n+2}}{6(3^n)} \quad \text{and} \quad b = \frac{3y_{n+1} - y_{n+2}}{-10(5^n)}$$

Substituting these values of a and b in (3) and simplifying, we get

$$y_{n+2} - 8y_{n+1} + 15y_n = 0,$$

which is required difference equation.

Example

Find the difference equations for the family of curves $y(x) = Ax + B2^x$, where A and B are unknowns.

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Solution

$$y_x = Ax + B2^x \tag{6}$$
$$\implies y_{x+1} = A(x+1) + B2^{x+1} \tag{7}$$
$$\implies y_{x+2} = A(x+2) + B2^{x+2} \tag{8}$$

Solving for A and B from (6) and (7) yields

$$A = \frac{y_{x+1} - 2y_x}{1-x} \quad \text{and} \quad B = \frac{1}{2^x(x-1)}\left[xy_{x+1} - (x+1)y_x\right]$$

Substituting these values of A and B in (8) and simplifying, we get

$$(1-x)y_{x+2} + (3x-2)y_{x+1} - 2xy_x = 0,$$

which is the required difference equation.

Exercise 1:

Obtain the difference equation for the sequence $\{y_n\}$, where

(a) $y_n = \frac{a}{n} + b$ **Ans.** $(n+2)y_{n+2} - 2(n+1)y_{n+1} + ny_n = 0.$

(b) $y_n = an^2 - bn$ **Ans.** $(n^2+n)y_{n+2} - (2n^2+4n)y_{n+1} + (n^2+3n+2)y_n = 0.$

(c) $y_n = a2^n + b3^n + c$ **Ans.** $y_{n+3} - 6y_{n+2} + 11y_{n+1} - 6y_n = 0.$

Exercise 2:

Form the difference equation generated by the family of curves given by $y_x = ax + b2^x$. **Ans.** $(x-1)y_{x+2} - (3x-2)y_{x+1} + 2xy_x = 0.$

Exercise 3:

Form the difference equation corresponding to the family of curves $y = ax + bx^2$.

Ans. $x(x+1)y_{x+2} - (2x^2+4x)y_{x+1} + (x^2+3x+2)y_x = 0.$

Definition(Linear difference equations with constant coefficients)

A linear difference equation is that in which $y_n, y_{n+1}, y_{n+2}, \ldots$ etc. occur to the first degree only and are not multiplied together.

If the coefficients $y_n, y_{n+1}, y_{n+2}, \ldots$ etc. are constants then the linear difference equation is called linear difference equation with constant coefficients. A linear difference equation with constant coefficients has the following form

$$y_{n+r} + a_1 y_{n+r-1} + a_2 y_{n+r-2} + \cdots + a_n y_n = f(n), \tag{9}$$

where $a_1, a_2, \ldots, a_n$ are constants. If $f(n) = 0$, then (9) is called Homogeneous difference equation otherwise it is called non-homogeneous or Inhomogeneous difference equation.

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Shifting operator E

Let $y = f(x)$ be given function of x. Let $y_0, y_1, y_2, \ldots, y_n$ be the values of y at $x = x_0, x_1, x_2, \ldots, x_n$. Then the shifting operator E is defined as

$$Ey_0 = y_1, \; Ey_1 = y_2, \ldots, Ey_{n-1} = y_n$$

$E[Ey_0] = E(y_1) = y_2$ and in general $E^n y_0 = y_n$.

Then (9) can be expressed in terms of shifting operator as following

$$E^r y_n + a_1 E^{r-1} y_n + a_2 E^{r-2} y_n + \cdots + a_n y_n = f(n)$$
$$\implies [E^r + a_1 E^{r-1} + a_2 E^{r-2} + \cdots + a_n] y_n = f(n) \tag{10}$$

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Now we shall deal with linear difference equations with constant coefficients only. Their properties are analogous to those of linear differential equations with constant coefficients.

Complementary function and Particular integral

If $u_1(n), u_2(n), \ldots, u_r(n)$ be r independent solution of the homogeneous difference equation

$$y_{n+r} + a_1 y_{n+r-1} + a_2 y_{n+r-2} + \cdots + a_n y_n = 0$$

then its complete solution is

$$U_n = c_1 u_1(n) + \cdots + c_r u_r(n)$$

where $c_1, c_2, \ldots, c_r$ are arbitrary constants. This U_n is called **complementary function (C.F.)** of difference equation (10).

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If V_n is a particular solution of (10), then the complete solution of (10) is $y_n = U_n + V_n$. The part V_n is called the **particular integral (P.I.)** of (10). Thus, the complete solution (C.S.) of (10) is

$$y_n = \text{C.F.} + \text{P.I.}$$

Rule to find the Complementary function

Consider the difference equation

$$y_{n+r} + a_1 y_{n+r-1} + a_2 y_{n+r-2} + \cdots + a_n y_n = 0, \tag{11}$$

where $a_1, a_2, \ldots, a_n$ are constants. Then this equation in symbolic form is

$$[E^r + a_1 E^{r-1} + a_2 E^{r-2} + \cdots + a_n] y_n = 0$$

The equation

$$m^r + a_1 m^{r-1} + a_2 m^{r-2} + \cdots + a_n = 0$$

is called **Auxiliary Equation (A.E.)** of difference equation (10). Since, it is a polynomial of degree r hence it has r roots say $\lambda_1, \lambda_2, \cdots, \lambda_r$.

Case-I If all the roots are real and distinct, then the General Solution of the difference equation (11) is given by

$$y_n = c_1 {\lambda_1}^n + c_2 {\lambda_2}^n + \cdots + c_r {\lambda_r}^n$$

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Case-II If two roots are equal (i .e. $\lambda_1 = \lambda_2$), then the General Solution of the difference equation (11) is given by

$$y_n = (c_1 + nc_2){\lambda_1}^n + \cdots + c_r {\lambda_r}^n$$

If, however, the A.E. has three equal roots (i .e. $\lambda_1 = \lambda_2 = \lambda_3$), then the General Solution of the difference equation (11) is given by

$$y_n = (c_1 + nc_2 + n^2 c_3){\lambda_1}^n + c_4 {\lambda_4}^n + \cdots + c_r {\lambda_r}^n$$

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ase-III If one pair of roots be imaginary, i .e. $\lambda_1 = \alpha + i\beta, \lambda_2 = \alpha - i\beta$, then the General Solution of the difference equation (11) is given by

$$y_n = R^n[c_1 \cos n\theta + c_2 \sin n\theta] + c_3 {\lambda_3}^n + \cdots + c_r {\lambda_r}^n,$$

where $R = \sqrt{\alpha^2 + \beta^2}$ and $\theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$.

Example

Solve the difference equation $y_{n+3} - 4y_{n+2} + y_{n+1} + 6y_n = 0$.

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Solution

The given difference equation can be written as

$$(E^3 - 4E^2 + E + 6)y_n = 0.$$

Its auxiliary equation is

$$m^3 - 4m^2 + m + 6 = 0 \implies (m+1)(m-2)(m-3) = 0$$

Thus, the roots of the auxiliary equation are $-1, 2, 3$, which are real and distinct. Hence the solution to given difference equation is

$$y_n = c_1(-1)^n + c_2 2^n + c_3 3^n.$$

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Solution

The given difference equation can be written in symbolic form as

$$(E^2 - 2E + 1)u_n = 0.$$

Its auxiliary equation is

$$m^2 - 2m + 1 = 0 \implies (m-1)^2 = 0$$

Thus, the roots of the auxiliary equation are $1, 1$, which are real and equal. Hence, the solution to given difference equation is

$$y_n = (c_1 + nc_2)1^n = c_1 + nc_2.$$

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Solution

The given difference equation can be written in symbolic form as

$$(E^2 + 2E + 4)u_n = 0.$$

Its auxiliary equation is

$$m^2 + 2m + 4 = 0 \implies m = -1 \pm \sqrt{3}i$$

Thus, the roots of the auxiliary equation are $-1 + \sqrt{3}i, -1 - \sqrt{3}i$, which are imaginary roots. Hence, the solution to given difference equation is

$$y_n = 2^n[c_1 \cos n\theta + c_2 \sin n\theta],$$

where $\theta = \tan^{-1}(-\sqrt{3}) = \frac{2\pi}{3}$. Thus,

$$y_n = 2^n \left[c_1 \cos\left(n\frac{2\pi}{3}\right) + c_2 \sin\left(n\frac{2\pi}{3}\right)\right].$$

Fibonacci sequence

Fibonacci Number: In the year 1202, an Italian mathematician Leonardo of Pisa, better known as Fibonacci, posed the following problem in his book Liber Abaci : A male and a female rabbit are born at the beginning of the year. We assume the following conditions:

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How many rabbits will there be at the end of the year?

Mathematically, Let y_n denote the number of rabbit pairs at the beginning of month n, then we have

$y_0 = 1,\ y_1 = 1,\ y_2 = 2,\ y_3 = 3,\ y_4 = 5, \ldots$ (Look at the diagram) and, in general,

$$y_n = y_{n-1} + y_{n-2}$$

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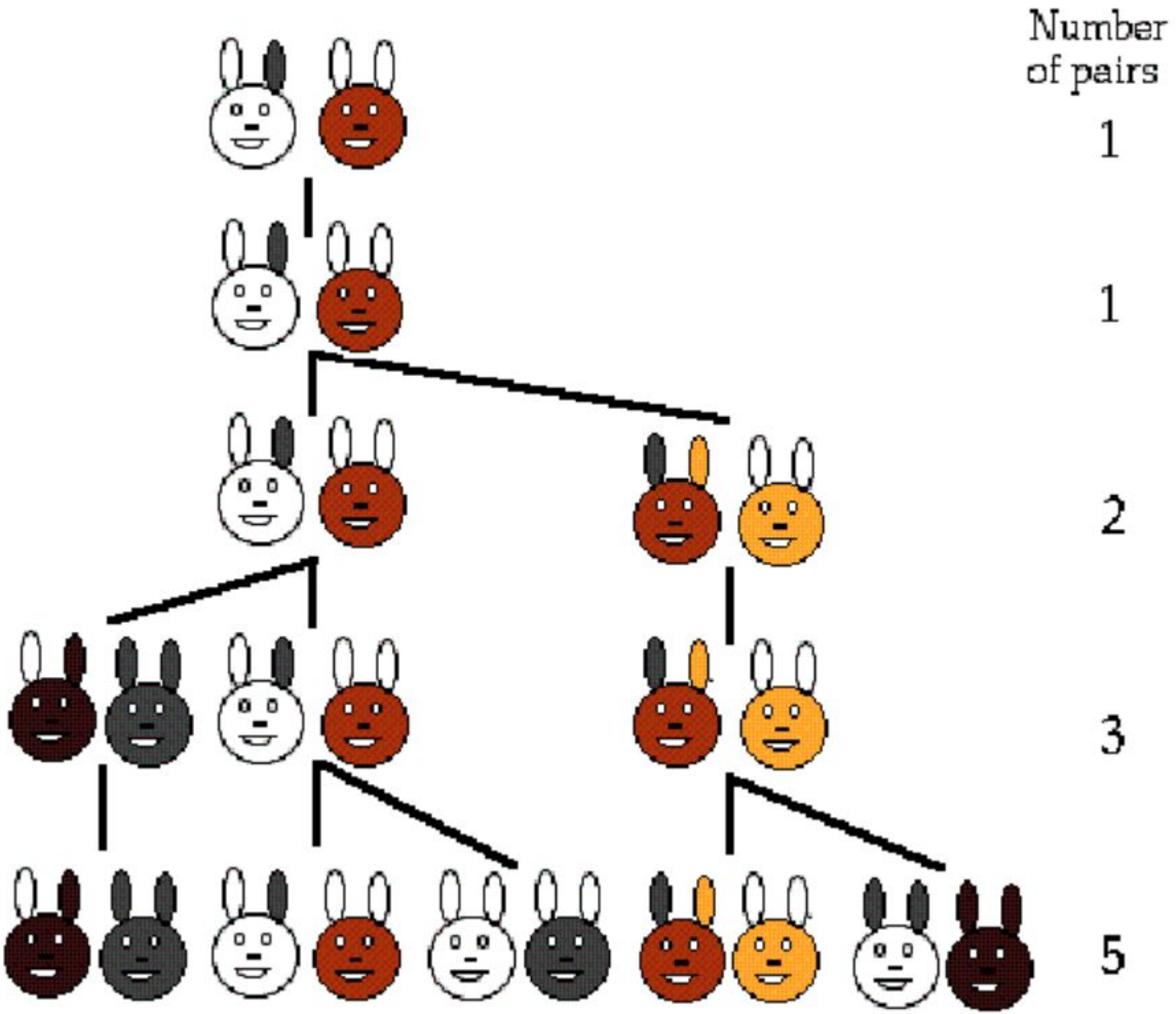
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$$y_n = y_{n-1} + y_{n-2}$$

• **Definition:** The integers $1, 1, 2, 3, 5, 8, 13, 21, \ldots$ are said to form a Fibonacci sequence satisfying the condition

$$y_n = y_{n-1} + y_{n-2} \qquad \text{for} \qquad n > 2.$$


Number
of pairs
1
1
2
3
5

The following table gives the first 55 Fibonacci numbers:

n, y_n	n, y_n	n, y_n	n, y_n	n, y_n
1, 1	**12, 144**	23, 28657	34, 5702887	45, 1134903170
2, 1	13, 233	24, 46368	35, 9227465	46, 1836311903
3, 2	14, 377	25, 75025	36, 14930352	47, 2971215073
4, 3	15, 610	26, 121393	37, 24157817	48, 4807526976
5, 5	16, 987	27, 196418	38, 39088169	49, 7778742049
6, 8	17, 1597	28, 317811	39, 63245986	50, 12586260925
7, 13	18, 2584	29, 514229	40, 102334155	51, 20365011074
8, 21	19, 4181	30, 832040	41, 165580141	52, 32951280099
9, 34	20, 6765	31, 1346269	42, 267914296	53, 53316291173
10, 55	21, 10946	32, 2178309	43, 433494437	54, 86267571272
11, 89	22, 17711	33, 3524578	44, 701408733	55, 139583862445

In particular, at the end of the first year (n=12), there will 144 rabbits.

Example

Solve the Fibonacci difference equation $y_n = y_{n-1} + y_{n-2}, \quad n > 2.$

Example

Solve the Fibonacci difference equation $y_n = y_{n-1} + y_{n-2}, \quad n > 2.$

Solution

The Fibonacci difference equation can be written as $y_{n+2} - y_{n+1} - y_n = 0, \quad n > 0.$
The given difference equation can be written in symbolic form as

$$(E^2 - E - 1)y_n = 0.$$

The auxiliary equation is

$$m^2 - m - 1 = 0 \implies m = \frac{1 \pm \sqrt{5}}{2}$$

Thus, the roots of the auxiliary equation are $\frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}$, which are imaginary roots.
Hence, the solution to given difference equation is

$$y_n = c_1 \left(\frac{1 + \sqrt{5}}{2}\right)^n + c_2 \left(\frac{1 - \sqrt{5}}{2}\right)^n$$

Solution cont...

when $n = 0$, $y_n = 1$. Therefore

$$c_1 + c_2 = 1 \tag{12}$$

when $n = 1$, $y_n = 1$. Therefore

$$c_1 \left(\frac{1+\sqrt{5}}{2} \right) + c_2 \left(\frac{1-\sqrt{5}}{2} \right) = 1 \tag{13}$$

Solving (12) and (13) for c_1 and c_2, we get

$$c_1 = \frac{5+\sqrt{5}}{10} = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right), \quad c_2 = \frac{5-\sqrt{5}}{10} = -\frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)$$

Therefore

$$y_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{n+1}$$

Exercise 4:

Solve the following difference equations

(a) $y_{n+3} - 2y_{n+2} - 5y_{n+1} + 6y_n = 0$

(b) $\Delta^2 y_n + 2\Delta y_n + y_n = 0$

(c) $u_{n+3} - 3u_{n+1} + 2u_n = 0$, where $u_1 = 0, u_2 = 8, u_3 = -2$

Exercise 5:

Solve the following difference equation

(a) $y_{k+2} - 7y_{k+1} + 6y_k = 0$ Ans: $y_k = c_1 6^k + c_2,$

(b) $y_{k+2} - y_k = 0$ Ans: $y_k = c_1 + c_2(-1)^k,$

(c) $y_{k+2} - 6y_{k+1} + 9y_k = 0$ Ans: $y_k = (c_1 + c_2 k)3^k,$

(d) $y_{k+2} = y_{k+1} + y_k$ Ans: $y_k = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^k + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^k$

(e) $y_{k+3} + y_{k+2} - 8y_{k+1} - 12y_k = 0$ Ans: $y_k = c_1 3^k + (c_2 + c_3 k)(-2)^k,$

(f) $y_{k+3} + y_{k+2} - y_{k+1} - y_k = 0$ Ans: $y_k = c_1 + (c_2 + c_3 k)(-1)^k.$

(g) $y_{k+2} - 2y_{k+1} + 2y_k = 0$ Ans: $y_k = c_1 2^{k/2} \cos\left(\frac{\pi k}{4}\right) + c_2 2^{k/2} \sin\left(\frac{\pi k}{4}\right)$

(h) $y_{k+4} - y_k = 0$ Ans: $y_k = c_1 \cos\left(\frac{\pi k}{2}\right) + c_2 \sin\left(\frac{\pi k}{2}\right) + c_3(-1)^k + c_4,$

(i) $y_{k+4} + y_k = 0.$ Ans: $y_k = c_1 \cos\left(\frac{\pi k}{4}\right) + c_2 \sin\left(\frac{\pi k}{4}\right) + c_3 \cos\left(\frac{\pi k}{4}\right) + c_4 \sin\left(\frac{\pi k}{4}\right)$

(j) $y_{k+1} - (2\cos\phi)y_k + y_{k-1} = 0,\ \cos\phi \neq 0,$ Ans: $y_k = c_1 \cos(\phi k) + c_2 \sin(\phi k),$

Rules for finding the Particular Integral

There are many ways for finding Particular Integral. Some of these are :

1. Operator Method
2. Method of Undetermined Coefficients

In your syllabus only last, i.e., **Method of Undetermined Coefficients** is there.

Method of Undetermined Coefficients

Consider the following non-homogeneous linear differential equation with constant coefficients

$$y_{n+2} + ay_{n+1} + by_n = Q(n) \tag{14}$$

where a and b are constants and $Q(n)$ is a function of n. Then

(1) A particular integral of (14) can be found by the method of undetermined coefficients whenever the non-homogeneous term $Q(n)$ is a linear combination of elementary functions such as polynomials and β^n for constant β.

(2) The method depends for its success on recognizing the general form of a function that when substituted into the left-hand side of (14) yields the general form of the non-homogeneous term $Q(n)$ on the right-hand side.

(3) Undetermined coefficients are involved because although the general form of a particular integral y_p can be guessed from the function $Q(n)$, any multiplicative constants (the undetermined coefficients) involved will not be known.

(4) Their values are found by substituting the possible form for y_p into the left-hand side of (14) and equating the undetermined coefficients of terms on the left of the equation to the known coefficients of corresponding terms in $Q(n)$ on the right.

Choice Rules for the Method of Undetermined Coefficients

(a) **Basic Rule.** If $Q(n)$ in (14) is one of the functions in the first column in Table 1, choose in the same line and determine its undetermined coefficients by substituting into (14).

(b) **Modification Rule.** If a term in your choice for happens to be a solution of the homogeneous ODE corresponding to (14), multiply this term by n (or by n^2 if this solution corresponds to a double root of the characteristic equation of the homogeneous ODE).

(c) **Sum Rule.** If $Q(n)$ is a sum of functions in the first column of Table 1, choose for the sum of the functions in the corresponding lines of the second column.

If $Q(n)$ has a term that is a constant multiple of...	**And if**	**Then include this expression in the trial function for y_p**
β^n	β is not a root of the auxiliary equation	$A\beta^n$
	β is a single root of the auxiliary equation	$An\beta^n$
	β is a double root of the auxiliary equation	$An^2\beta^n$
pn^2+qn+r	1 is not a root of the auxiliary equation	$An^2 + Bn + C$
	1 is a single root of the auxiliary equation	$An^3 + Bn^2 + Cn$
	1 is a double root of the auxiliary equation	$An^4 + Bn^3 + Cn^2$
$p(n)\beta^n$, here $p(n)$ is a polynomial of the degree r in n and β is a constant	β is not a root of the auxiliary equation	$[a_0 + a_1 n + \cdots + a_r n^r]\beta^n$
	β is a single root of the auxiliary equation	$n[a_0 + a_1 n + \cdots + a_r n^r]\beta^n$
	β is a double root of the auxiliary equation	$n^2[a_0 + a_1 n + \cdots + a_r n^r]\beta^n$
$\cos \beta n$ or $\sin \beta n$	If $\cos \beta n$ and $\sin \beta n$ do not appear in the complementary function	$A \cos \beta n + B \sin \beta n$
	If $\cos \beta n$ or $\sin \beta n$ appear in the complementary function	$An \cos \beta n + Bn \sin \beta n$

Table 1: The method of undetermined coefficients

Example

Solve the difference equation $y_{n+2} + 5y_{n+1} + 6y_n = 3n^2$.

Example

Solve the difference equation $y_{n+2} + 5y_{n+1} + 6y_n = 3n^2$.

Solution

The given difference equation can be written in symbolic form as

$$(E^2 + 5E + 6)y_n = 3n^2$$

Its auxiliary equation is

$$m^2 + 5m + 6 = 0 \implies (m+2)(m+3) = 0$$

Thus, the roots of the auxiliary eqution are $-2, -3$, which are real and distinct. Hence, the complementary function of given difference equation is

$$y_h = c_1(-2)^n + c_2(-3)^n$$

Let the particular integral of given difference equation be

$$V_n = An^2 + Bn + C$$

Solution cont...

This gives

$$V_{n+1} = A(n+1)^2 + B(n+1) + C \quad \text{and} \quad V_{n+2} = A(n+2)^2 + B(n+2) + C$$

Since, V_n is a solution of the given differential equation. Therefore

$$\begin{aligned} & V_{n+2} + 5V_{n+1} + 6V_n = 3n^2 \\ \implies & A(n+2)^2 + B(n+2) + C + 5[A(n+1)^2 + B(n+1) + C] + 6[An^2 + Bn + C] = 3n^2 \\ \implies & 12An^2 + (14A + 12B)n + (9A + 7B + 12C) = 3n^2 \end{aligned}$$

Comparing the respective coefficients, we get

$$A = \frac{3}{12} = \frac{1}{4}, 14A + 12B = 0 \implies B = -\frac{7}{24}, 9A + 7B + 12C = 0 \implies C = -\frac{5}{288}$$

So the particular solution is

$$V_n = \frac{1}{4}n^2 - \frac{7}{24}n - \frac{5}{288}$$

Hence the general solution of the given difference equation

$$y_n = c_1(-2)^n + c_2(-3)^n + \frac{1}{4}n^2 - \frac{7}{24}n - \frac{5}{288}$$

Example

Solve the difference equation $y_{n+2} - 6y_{n+1} + 5y_n = 2^n$ with $y_0 = 0$, $y_1 = 0$.

Example

Solve the difference equation $y_{n+2} - 6y_{n+1} + 5y_n = 2^n$ with $y_0 = 0$, $y_1 = 0$.

Solution

The auxiliary equation of the given difference equation is

$$m^2 - 6m + 5 = 0 \implies (m-1)(m-5) = 0$$

Thus, the roots of the auxiliary equations are $1, 5$, which are real and distinct. Hence, the complementary function is given by

$$y_h = c_1(1)^n + c_2 5^n = c_1 + c_2 5^n$$

Let the particular integral of the given difference equation be

$$V_n = A2^n$$

This gives

$$V_{n+1} = A2^{n+1} \quad \text{and} \quad V_{n+2} = A2^{n+2}$$

Since V_n is a solution of the given difference equation. Therefore

$$\begin{aligned} & V_{n+2} - 6V_{n+1} + 5V_n = 2^n \\ \implies & A2^{n+2} - 6A2^{n+1} + 5A2^n = 2^n \\ \implies & -3A2^n = 2^n \implies A = -\frac{1}{3} \end{aligned}$$

So, the particular integral is

$$V_n = -\frac{1}{3}2^n$$

Hence, the general solution of the given difference equation is given by

$$y_n = c_1 + c_2 5^n - \frac{1}{3}2^n$$

Since $y_0 = 0$ and $y_1 = 0$ therefore

$$c_1 + c_2 = \frac{1}{3} \qquad\qquad c_1 + 5c_2 = \frac{2}{3}$$

Solving $c_1 = \frac{1}{4}, c_2 = \frac{1}{12}$. Then

$$y_n = \frac{1}{4} + \frac{1}{12}5^n - \frac{1}{3}2^n$$

Example

Solve the recurrence relation $y_{n+2} - 4y_{n+1} + 4y_n = 2^n$.

Example

Solve the recurrence relation $y_{n+2} - 4y_{n+1} + 4y_n = 2^n$.

Solution

The auxiliary equation of the given recurrence relation is given by

$$m^2 - 4m + 4 = 0 \implies (m-2)^2 = 0,$$

which has two real equal roots $2, 2$. Therefore the complementary function is given by

$$y_h = (c_1 + c_2 n)2^n$$

Since 2 is a double root of the auxiliary equation and hence let the particular integral be

$$V_n = An^2 2^n$$

This gives

$$V_{n+1} = A(n+1)^2 2^{n+1} \quad \text{and} \quad V_{n+2} = A(n+2)^2 2^{n+2}$$

Solution cont...

Since V_n is the solution of given recurrence relation and hence

$$\begin{aligned} & A(n+2)^2 2^{n+2} - 4A(n+1)^2 2^{n+1} + 4An^2 2^n = 2^n \\ \implies & 4A(n^2+4n+4)2^n - 8A(n^2+2n+1)2^n + 4An^2 2^n = 2^n \\ \implies & 12An^2 2^n = 2^n \\ \implies & A = \frac{1}{12} \end{aligned}$$

So, the particular integral is

$$V_n = \frac{1}{12} n^2 2^n$$

Therefore the general solution of the given recurrence relation is

$$y_n = (c_1 + c_2 n)2^n + \frac{1}{12} n^2 2^n$$

Example

Find the particular integral of the following recurrence relation

$$y_{n+2} - 5y_{n+1} + 4y_n = n$$

Example

Find the particular integral of the following recurrence relation

$$y_{n+2} - 5y_{n+1} + 4y_n = n$$

Solution

Since 1 is a root of the auxiliary equation of the given difference equation, therefore let the particular integral be

$$V_n = An^2 + Bn,$$

which gives

$$V_{n+2} = A(n+2)^2 + B(n+2) \quad \text{and} \quad V_{n+1} = A(n+1)^2 + B(n+1)$$

Since V_n is a solution of given recurrence relation, so

$$\begin{aligned} & A(n+2)^2 + B(n+2) - 5[A(n+1)^2 + B(n+1)] + 4[An^2 + Bn] = n \\ \Longrightarrow\ & (4A + B - 10A - B + 4B)n + (4A + 2B - 5A - B) = n \\ \Longrightarrow\ & (-6A + 4B)n + (-A + B) = n \end{aligned}$$

Solution cont. . .

Comparing the coefficients of respective terms, we get

$$-6A + 4B = 1 \qquad\qquad -A + B = 0$$

Solving these two equations, we have

$$A = B = -\frac{1}{2}$$

Therefore the particular integral of given difference equation is

$$V_n = -\frac{1}{2}(n^2 + n)$$

Example

Find the general solution of the following difference equation

$$y_{n+2} - 3y_{n+1} + 2y_n = 2\sin(3n)$$

Example

Find the general solution of the following difference equation

$$y_{n+2} - 3y_{n+1} + 2y_n = 2\sin(3n)$$

Solution

The complementary function of the given difference equation is given by

$$y_n{}^{(H)} = c_1 + c_2 2^n,$$

which do not contain $\sin 3n$ or $\cos 3n$. Therefore, the particular solution to given difference equation can be written

$$y_n{}^{(P)} = A\sin(3n) + B\cos(3n).$$

This gives

$$\begin{aligned} y_{n+1}{}^{(P)} &= A\sin 3(n+1) + B\cos 3(n+1) \\ &= A[\sin 3n\cos 3 + \cos 3n\sin 3] + B[\cos 3n\cos 3 - \sin 3n\sin 3] \\ &= \sin 3n[A\cos 3 - B\sin 3] + \cos 3n[A\sin 3 + B\cos 3] \end{aligned}$$

Solution cont. . .

and

$$\begin{aligned} {y_{n+2}}^{(P)} &= A\sin 3(n+2) + B\cos 3(n+2) \\ &= A[\sin 3n\cos 6 + \cos 3n\sin 6] + B[\cos 3n\cos 6 - \sin 3n\sin 6] \\ &= \sin 3n[A\cos 6 - B\sin 6] + \cos 3n[A\sin 6 + B\cos 6] \end{aligned}$$

Since ${y_n}^{(P)}$ is a solution of the given difference equation and therefore

$$\begin{aligned} & {y_{n+2}}^{p} - 3{y_{n+1}}^{p} + 2{y_n}^{p} = 2\sin(3n) \\ \Longrightarrow\ & \sin 3n[A\cos 6 - B\sin 6] + \cos 3n[A\sin 6 + B\cos 6] - \\ & \qquad 3(\sin 3n[A\cos 3 - B\sin 3] + \cos 3n[A\sin 3 + B\cos 3]) + \\ & \qquad 2(A\sin(3n) + B\cos(3n)) = 2\sin(3n) \\ \Longrightarrow\ & \{A[\cos(6) - 3\cos(3) + 2] + B[-\sin(6) + 3\sin(3)]\}\sin(3n) + \\ & \qquad \{A[\sin(6) - 3\sin(3)] + B[\cos(6) - 3\cos(3) + 2]\}\cos(3n) = 2\sin(3n). \end{aligned}$$

Comparing the coefficients of $\sin 3n$ and $\cos 3n$, we get

$$\begin{aligned} A[\cos(6) - 3\cos(3) + 2] + B[-\sin(6) + 3\sin(3)] &= 2, \\ A[\sin(6) + 3\sin(3)] + B[\cos(6) - 3\cos(3) + 2] &= 0. \end{aligned}$$

Solution cont. . .

Solving above two equations for A and B, we get

$$A = 0.337, \ B = 0.039,$$

Therefore

$$y_n{}^{(P)} = 0.337 \sin(3n) + 0.039 \cos(3n).$$

Thus, the general solution of the given difference equation is

$$y_n = y_n{}^{(H)} + y_n{}^{(P)} = c_1 + c_2 2^n + 0.337 \sin(3n) + 0.039 \cos(3n)$$

Exercise 6:

Solve the following recurrence relation using method of undetermined coefficients

(a) $y_{n+1} + y_n = 3n2^n$.

(b) $y_{n+2} - 4y_{n+1} + 4y_n = (n+1)2^n$.

(c) $y_{n+2} - 5y_{n+1} + 6y_n = n + 2^n$.

Exercise 7:

Solve the following difference equation using method of undetermined coefficients

(a) $y_{k+2} - 5y_{k+1} + 6y_k = 2 + 4k$ Ans: $y_k = c_1 3^k + c_2 2^k + 4 + 2k$

(b) $y_{k+2} - 6y_{k+1} + 8y_k = 2 + 3k^2 - 5 \cdot 3^k$

Ans: $y_k = c_1 2^k + c_2 4^k + 44/9 + 8/3k + k^2 + 5 \cdot 3^k$.

(c) $y_{k+2} - 4y_{k+1} + 3y_k = k4^k$ Ans: $y_k = c_1 + c_2 3^k - \frac{1}{9}(16 - 3k)4^k$

(d) $y_{k+3} - 7y_{k+2} + 16y_{k+1} - 12y_k = k2^k$.

Ans: $y_k = (c_1 + c_2 k)2^k + c_3 3^k - \frac{1}{24}(3 + k)k^2 2^k$.

Method of Z-Transforms

Z-transforms can also be used to find the solution of difference equations. The following steps are required to solve a difference equation using method of Z-transform:

Step 1. Take Z-transform on both sides of the given difference equation.

Step 2. Use given conditions and solve for $U(z)$.

Step 3. Apply partial fractions method.

Step 4. Take inverse Z-transform on both sides which results in the given sequence.

Example

(a) Using the Z-transform, solve the difference equation $y_{n+1} + 3y_n = 0$, $y_0 = 1$.

(b) Solve the difference equation

$$y_{k+2} + 4y_{k+1} + 3y_k = 3^k,$$

given that $y_0 = 0, y_1 = 1$.

Solution

(a) Taking Z-transform on both sides, we get

$$Z(y_{n+1}) + 3Z(y_n) = Z(0) = 0$$

Denote $Z(y_n)$ by $Y(z)$, then

$$Z(y_{n+1}) = z[Y(z) - y_0] = z[Y(z) - 1]$$

Substituting these values

$$z[Y(z) - 1] + 3Y(z) = 0 \implies Y(z) = \frac{z}{z+3}$$

Taking inverse Z-transform, we get

$$Z^{-1}(Y(z)) = Z^{-1}\left(\frac{z}{z+3}\right) \implies y_n = (-3)^n$$

Solution cont...

(b) Taking Z-transform on both sides, we get

$$Z(y_{k+2}) + 4Z(y_{k+1}) + 3Z(y_k) = Z(3^k)$$

Denote $Z(y_k)$ by $Y(z)$, then

$$Z(y_{k+1}) = z[Y(z) - y_0] = zY(z)$$
$$Z(y_{k+2}) = z^2[Y(z) - y_0 - \tfrac{y_1}{z}] = z^2[Y(z) - \tfrac{1}{z}]$$

Substituting these values

$$z^2\left[Y(z) - \frac{1}{z}\right] + 4zY(z) + 3Y(z) = \frac{z}{z-3}$$

$$\implies (z^2 + 4z + 3))Y(z) = \frac{z}{z-3} + z$$

$$\implies \frac{Y(z)}{z} = \frac{1}{(z+1)(z+3)(z-3)} + \frac{1}{(z+1)(z+3)}$$

$$\implies \frac{Y(z)}{z} = \frac{z-2}{(z+1)(z+3)(z-3)}$$

Solution cont...

Breaking it into partial fractions, we get

$$\frac{Y(z)}{z} = \frac{1}{24(z-3)} - \frac{5}{12(z+3)} + \frac{3}{8(z+1)}$$

$$\implies Y(z) = \frac{1}{24}\left(\frac{1}{z-3}\right) - \frac{5}{12}\left(\frac{1}{z+3}\right) + \frac{3}{8}\left(\frac{1}{z+1}\right)$$

Taking inverse Z-transform of both sides yields

$$Z^{-1}(Y(z)) = \frac{1}{24}Z^{-1}\left(\frac{1}{z-3}\right) - \frac{5}{12}Z^{-1}\left(\frac{1}{z+3}\right) + \frac{3}{8}Z^{-1}\left(\frac{1}{z+1}\right)$$

$$\implies y_k = \frac{1}{24}3^k - \frac{5}{12}(-3)^k + \frac{3}{8}(-1)^k,$$

which is the required solution of the given difference equation.

Example

Using Z-transform, solve the difference equation $y_{n+2} - 4y_{n+1} + 3y_n = 5^n$ with $y_0 = y_1 = 1$.

Solution

Taking Z-transform on both sides of given difference equation yields

$$Z(y_{n+2}) - 4Z(y_{n+1}) + 3Z(y_n) = Z(5^n)$$

Denote $Z(y_n)$ by $Y(z)$, then

$$Z(y_{n+1}) = z[Y(z) - y_0] = z[Y(z) - 1]$$
$$Z(y_{n+2}) = z^2[Y(z) - y_0 - \tfrac{y_1}{z}] = z^2[Y(z) - 1 - \tfrac{1}{z}]$$

Substituting these values

$$z^2[Y(z) - 1 - \frac{1}{z}] - 4z[Y(z) - 1] + 3Y(z) = \frac{z}{z-5}$$
$$(z^2 - 4z + 3)Y(z) - (z^2 - 3z) = \frac{z}{z-5}$$
$$Y(z) = \frac{z}{(z-5)(z-3)(z-1)} + \frac{z^2 - 3z}{(z-1)(z-3)}$$
$$\frac{Y(z)}{z} = \frac{1}{(z-5)(z-3)(z-1)} + \frac{1}{(z-1)}$$

Breaking into partial fraction, we get

Solution cont...

$$\frac{Y(z)}{z} = \frac{9}{8}\frac{1}{z-1} + \frac{1}{8}\frac{1}{z-5} - \frac{1}{4}\frac{1}{z-3}$$

$$\implies Y(z) = \frac{9}{8}\left(\frac{z}{z-1}\right) + \frac{1}{8}\left(\frac{z}{z-5}\right) - \frac{1}{4}\left(\frac{z}{z-3}\right)$$

Taking inverse Z-transform of both sides, we get

$$Z^{-1}(Y(z)) = \frac{9}{8}Z^{-1}\left(\frac{z}{z-1}\right) + \frac{1}{8}Z^{-1}\left(\frac{z}{z-5}\right) - \frac{1}{4}Z^{-1}\left(\frac{z}{z-3}\right)$$

$$\implies y_n = \frac{9}{8} + \frac{1}{8}5^n - \frac{1}{4}3^n,$$

which is the required solution of the given difference equation.

Example

Solve the following difference equation using Z-transform

$$y_{n+2} - 2y_{n+1} + y_n = n, y_0 = 1, y_1 = 1$$

Example

Solve the following difference equation using Z-transform

$$y_{n+2} - 2y_{n+1} + y_n = n, y_0 = 1, y_1 = 1$$

Solution

Taking Z-transform on both sides of the given difference equation, we get

$$Z(y_{n+2}) - 2Z(y_{n+1}) + Z(y_n) = Z(n)$$

Denote $Z(y_n)$ by $Y(z)$, then

$$Z(y_{n+1}) = z[Y(z) - y_0] = z[Y(z) - 1]$$
$$Z(y_{n+2}) = z^2[Y(z) - y_0 - \tfrac{y_1}{z}] = z^2[Y(z) - 1 - \tfrac{1}{z}]$$

Substituting these values

$$z^2[Y(z) - 1 - \frac{1}{z}] - 2z[Y(z) - 1] + Y(z) = \frac{z}{(z-1)^2}$$

$$\implies Y(z) = \frac{z^4 - 3z^3 + 3z^2}{(z-1)^4}$$

Solution cont...

Now, Since

$$Z(1) = \frac{z}{z-1}, Z(n) = \frac{z}{(z-1)^2}, Z(n^2) = \frac{z^2+z}{(z-1)^3}, \text{ and } Z(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}$$

Therefore, let

$$Y(z) = \frac{z^4-3z^3+3z^2}{(z-1)^4} = \frac{A(z^3+4z^2+z)}{(z-1)^4} + \frac{B(z^2+z)}{(z-1)^3} + \frac{Cz}{(z-1)^2} + \frac{Dz}{z-1}$$

$$\implies z^4-3z^3+3z^2 = A(z^3+4z^2+z) + B(z^2+z)(z-1) + cz(z-1)^2 + Dz(z-1)^3$$

Setting $z = 1$, we get $A = \frac{1}{6}$. On comparing the coefficients of z^4, z^3, z^2 on both sides, we get $C = \frac{1}{3}$ and $B = -\frac{1}{2}$, $D = 1$. Thus,

$$Y(z) = \frac{1}{6}\left(\frac{z^3+4z^2+z}{(z-1)^4}\right) - \frac{1}{2}\left(\frac{z^2+z}{(z-1)^3}\right) + \frac{1}{3}\left(\frac{z}{(z-1)^2}\right) + \left(\frac{z}{z-1}\right)$$

Now, taking inverse Z-transform, we get

$$y_n = \frac{1}{6}n^3 - \frac{1}{2}n^2 + \frac{n}{3} + 1$$

Example

Solve the following difference equation using Z-transform method.

(a) $y_{k+2} = \frac{5}{6}y_{k+1} - \frac{1}{6}y_k + 3^k,\ y_0 = 0,\ y_1 = 1$

(b) $y_{k+2} + 6y_{k+1} + 9y_k = 2^k, y_0 = y_1 = 0.$

Solution

(a) Taking Z-transform of both sides of given difference equation, we get

$$Z(y_{k+2}) = \frac{5}{6}Z(y_{k+1}) - \frac{1}{6}Z(y_k) + Z(3^k)$$

Denote $Z(y_k)$ by $Y(z)$, then

$$Z(y_{k+1}) = z[Y(z) - y_0] = zY(z)$$
$$Z(y_{k+2}) = z^2[Y(z) - y_0 - \tfrac{y_1}{z}] = z^2[Y(z) - \tfrac{1}{z}]$$

Substituting these values

$$z^2\left[Y(z) - \frac{1}{z}\right] = \frac{5}{6}zY(z) - \frac{1}{6}Y(z) + \frac{z}{z-3}$$

$$\implies \frac{Y(z)}{z} = \frac{z^2 - 2z}{(z-3)(z-1/3)(z-1/2)}$$

Decomposing the R.H.S. into partial fraction we get

$$\frac{Y(z)}{z} = \frac{3}{20}\frac{1}{z-3} - \frac{15}{4}\frac{1}{z-1/3} + \frac{18}{5}\frac{1}{z-1/2}$$

$$\implies Y(z) = \frac{3}{20}\frac{z}{z-3} - \frac{15}{4}\frac{z}{z-1/3} + \frac{18}{5}\frac{z}{z-1/2}$$

Taking inverse Z-transform on both sides, gives

$$y_k = \frac{3}{20}\,3^k - \frac{15}{4}\frac{1}{3^k} + \frac{18}{5}\frac{1}{2^k}$$

(b) Taking Z-transform of both sides of given difference equation, we get

$$Z(y_{k+2}) + 6Z(y_{k+1}) + 9Z(y_k) = Z(2^k)$$

Denote $Z(y_k)$ by $Y(z)$, then

$$Z(y_{k+1}) = z[Y(z) - y_0] = zY(z)$$
$$Z(y_{k+2}) = z^2[Y(z) - y_0 - \tfrac{y_1}{z}] = z^2 Y(z)$$

Substituting these values

$$z^2 Y(z) + 6zY(z) + 9Y(z) = \frac{z}{z-2}$$

$$\implies \frac{Y(z)}{Z} = \frac{1}{(z-2)(z+3)^2}$$

Decomposing RHS into partial fraction yields

$$\frac{Y(z)}{z} = \frac{1}{25}\frac{1}{z-2} - \frac{1}{25}\frac{1}{z+3} - \frac{1}{5}\frac{1}{(z+3)^2}$$

$$\implies Y(z) = \frac{1}{25}\frac{z}{z-2} - \frac{1}{25}\frac{z}{z+3} - \frac{1}{5}\frac{z}{(z+3)^2}$$

$$\implies Y(z) = \frac{1}{25}\frac{z}{z-2} - \frac{1}{25}\frac{z}{z+3} + \frac{1}{15}\frac{(-3z)}{(z+3)^2}$$

Taking inverse Z-transform both sides, we get

$$y_k = \frac{1}{25}2^k - \frac{1}{25}(-3)^k + \frac{1}{15}k(-3)^k$$

Exercise 8:

Solve the following difference equation using Z-transform.

(a) $y_{n+1} - 5y_n = 0$ Ans: $y_n = y_0 5^n$, where a_0 is arbitrary

(b) $y_{n+2} - 3y_{n+1} + 2y_n = 0,\ y_0 = -1,\ y_1 = 2$ Ans: $y_n = 3(2^n) - 4$

(c) $y_{k+2} + 5y_{k+1}4y_k = 2^k,\ y_0 = \text{`},\ y_1 = -4$

Ans: $y_k = \frac{19}{18}(-4)^k - \frac{1}{9}(-1)^k + \frac{1}{18}(2^k)$

(d) $y_{k+2} - 6y_{k+1} + 9y_k = 3^k,\ y_0 = 0,\ y_1 = 1$ Ans: $y_n = \frac{1}{18}(5n + n^2)3^n$

(e) $y_{n+2} + 10y_{n+1} + 25y_n = n,\ y_0 = 1, y_1 = -5$

Ans: $y_n = \frac{1}{540}[(545 - 3n)(-5)^n + 5(3n - 1)]$

(f) $y_{n+2} + 4y_{n+1} + 3y_n = 2^n,\ y_0 = 0,\ y_1 = 1$ Ans: $y_n = \frac{(-1)^n}{3} + \frac{2^n}{15} - \frac{2}{5}(-3)^n$

(g) $y_{n+2} - 4y_{n+1} + 4y_n = 2^n,\ y_0 = 0, y_1 = 1$ Ans: $y_n = n2^{n-1} + n(n-1)2^{n-3}$

(h) $y_{n+2} - 2y_{n+1} + y_n = 3n + 5,\ y_0 = y_1 = 0$ Ans: $y_n = \frac{n(n-1)(n+3)}{2}$

(i) $y_{n+3} - 9y_{n+2} + 27y_{n+1} - 27y_n = 2^n,\ y - 0 = -1,\ y_1 = -3,\ y_2 = 2.$

Ans: $y_n = \frac{(2n^2-3n)3^n - 3(2)^n}{3}$

Finance: Compound interest

Example

If Raju invests Rs 1000 at 6% interest compounded quarterly, how many month must he wait for his money to double (Note that Raju can not withdraw the money before the quarter is up). How many months it trebles.

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If Raju invests Rs 1000 at 6% interest compounded quarterly, how many month must he wait for his money to double (Note that Raju can not withdraw the money before the quarter is up). How many months it trebles.

Solution

Annual interest rate is 6% so the quarterly rate is

$$\frac{6\%}{4} = \frac{3}{2}\% = \frac{3}{200} = 0.015.$$

For $0 \leq n \leq 4$, P_n denotes the value of Raju's deposit at the end of n quarters. Then $P_{n+1} = P_n + 0.015P_n$ where $0.015P_n$ is the interest earned on P_n during $(n+1)$th quarter. Here $P_0 = 1000$. The solution of the difference equation $P_{n+1} - 1.015Pn = 0$ is

$$P_n = P_0(1.015)^n = 1000(1.015)^n.$$

Solution cont. . .

If the money doubles then $P_n = 2P_0 = 2000$. Then

$$2000 = 1000(1.015)^n \implies n = \frac{\ln 2}{\ln(1.015)}$$

$$\implies n = 46.56 \approx 47 \text{ quarters.}$$

So money doubles in $47 \times 3 = 141$ months.
If money trebles then

$$3000 = 1000(1.015)^n \implies n = \frac{\ln 3}{\ln(1.015)} = 73.80 \approx 74,$$

$$74 \times 3 = 222.$$

Thus, money trebles in 222 months.

Application to Deflection of a Loaded String

Consider a light string of length l stretched tightly between A and B . Let the forces P_1 be acting at its equispaced points x_i $(i = 1, 2, \ldots, n-1)$ and perpendicular to AB resulting in small transverse displacements y_i; at these points (Fig. 1). Assuming the angle θ_i made by the portion between x_i and x_{i+1} with the horizontal, to be small, we have

$$\sin\theta_i \approx \theta_i \approx \tan\theta_i \text{ and } \cos\theta_i = 1$$

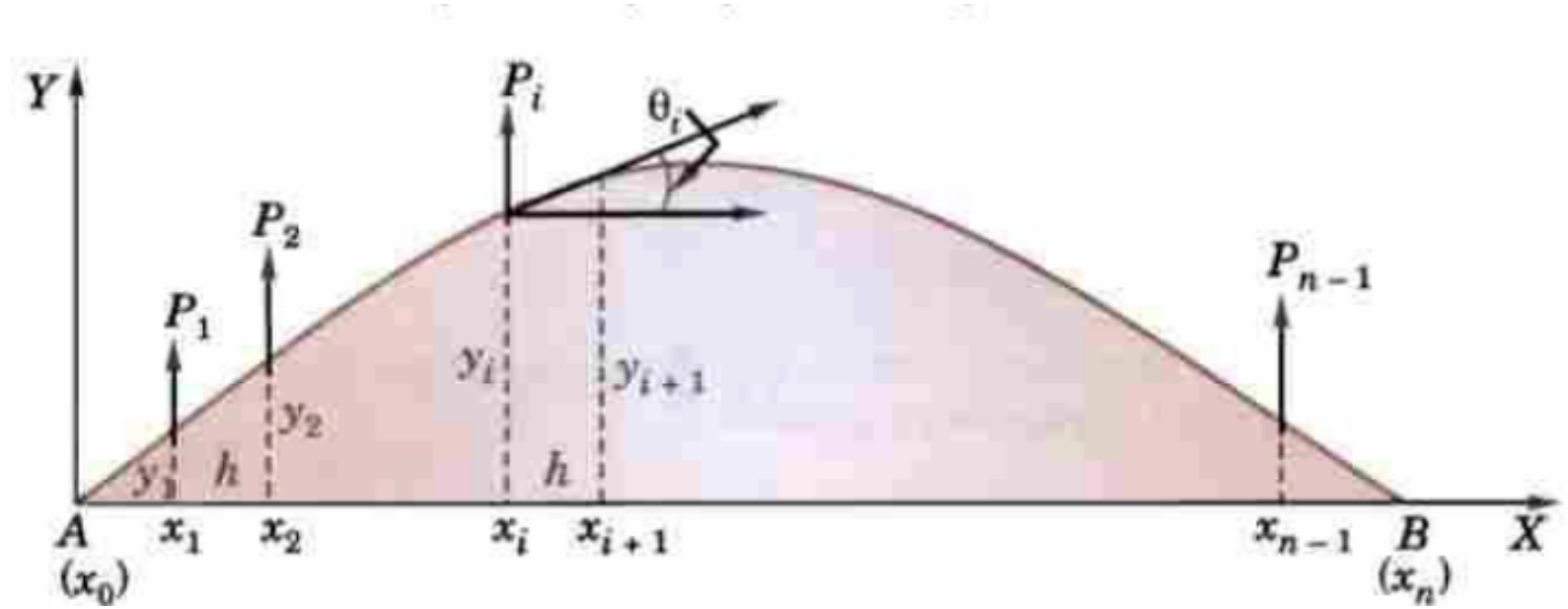


Figure 1: Deflection of a Loaded String

Cont. . .

If T be the tension of the string at x_i, then $T\cos\theta_i = T$, i.e., the tension may be taken as uniform. Taking $x_{i+1} - x_i = h$, we have

$$y_{i+1} - y_i = h\tan\theta_i = h\theta_i \tag{15}$$

$$y_i - y_{i-1} = h\tan\theta_{i-1} = h\theta_{i-1} \tag{16}$$

Also, resolving the forces in equilibrium at (x_i, y_i) perpendicular to AB, we get

$$T\sin\theta_i - T\sin\theta_{i-1} + P_i = 0 \implies T(\theta_i - \theta_{i-1}) + P_i = 0 \tag{17}$$

Eliminating θ_i and θ_{i-1} from (15), (16) and (17), we obtain

$$y_{i+1} - 2y_i + y_{i-1} = -\frac{hP_i}{T} \tag{18}$$

which is a difference equation and its solution gives the displacements y_i. To obtain the arbitrary constants in the solution, we take $y_0 = y_n = 0$ as the boundary conditions, since the ends A and B of the string are fixed.

Example

A light wing stretched between two fixed nails 120 cm apart, carries 11 loads of weight 5 gm each at equal intervals and the resulting tension is 500 gm weight. Show that the sag at the mid point is 1.8cm.

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Solution

Taking $h = 10cm$, $P_i = 5gm$ and $T = 500gm$ wt., the above equation (18) becomes

$$y_{i+1} - 2y_i + y_{i-1} = -\frac{10 \times 5}{500} = -\frac{1}{10}$$

The solution of above difference equation is given by

$$y_i = c_1 + c_2 i + \frac{1}{20}(i - i^2)$$

$$\because y_0 = 0 \implies c_1 = 0 \quad \text{and} \quad \because y_{12} = 0 \implies c_2 = \frac{11}{20}$$

Hence

$$y_i = \frac{11}{20} i + \frac{1}{20}(i - i^2)$$